

Heavy metals in commercial fish in New Jersey.

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Levels of contaminants in fish are of particular interest because of the potential risk to humans who consume them. While attention has focused on self-caught fish, most of the fish eaten by the American public comes from commercial sources. We sampled 11 types of fish and shellfish obtained from supermarkets and specialty fish markets in New Jersey and analyzed them for arsenic, cadmium, chromium, lead, manganese, mercury, and selenium. We test the null hypothesis that metal levels do not vary among fish types, and we consider whether the levels of any metals could harm the fish themselves or their predators or pose a health risk for human consumers. There were significant interspecific differences for all metals, and no fish types had the highest levels of more than two metals. There were few significant correlations (Kendall tau) among metals for the three most numerous fish (yellowfin tuna, bluefish, and flounder), the correlations were generally low (below 0.40), and many correlations were negative. Only manganese and lead positively were correlated for tuna, bluefish, and flounder. The levels of most metals were below those known to cause adverse effects in the fish themselves. However, the levels of arsenic, lead, mercury, and selenium in some fish were in the range known to cause some sublethal effects in sensitive predatory birds and mammals and in some fish exceeded health-based standards. The greatest risk from different metals resided in different fish; the species of fish with the highest levels of a given metal sometimes exceeded the human health guidance or standards for that metal. Thus, the risk information given to the public (mainly about mercury) does not present a complete picture. The potential of harm from other metals suggests that people not only should eat smaller quantities of fish known to accumulate mercury but also should eat a diversity of fish to avoid consuming unhealthy quantities of other heavy metals. However, consumers should bear in mind that standards have a margin of safety.